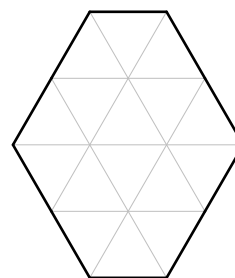
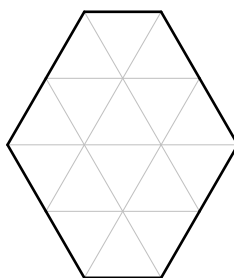
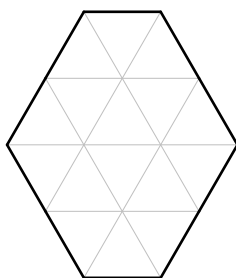
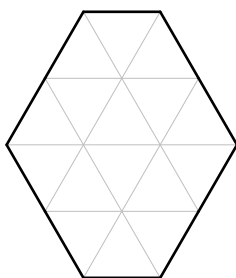
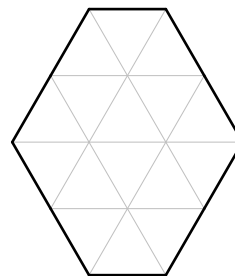
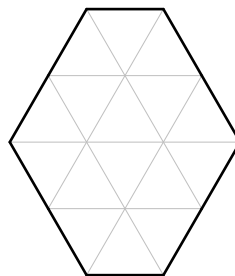
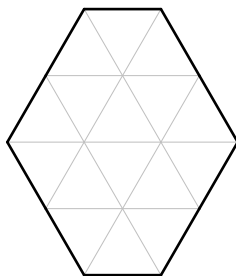
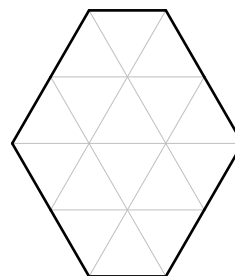
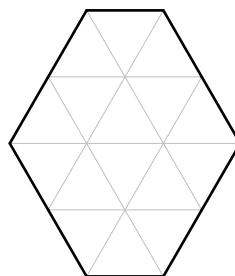
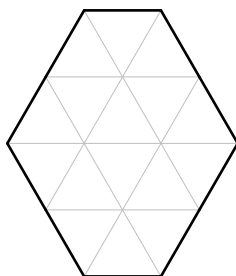
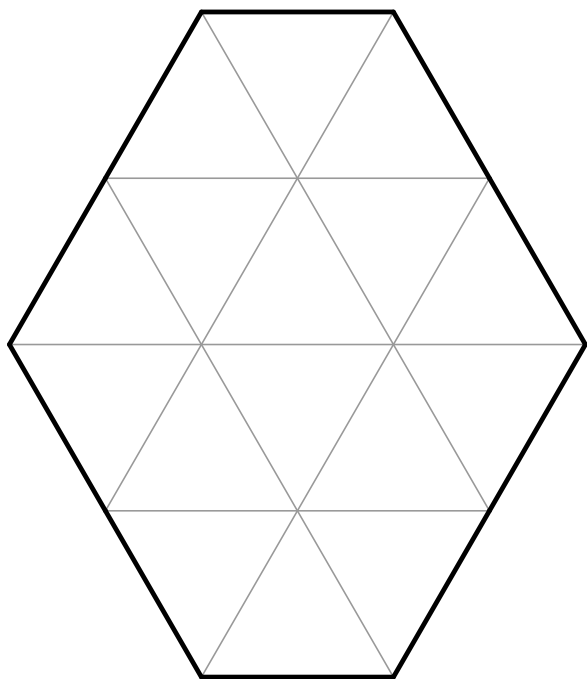
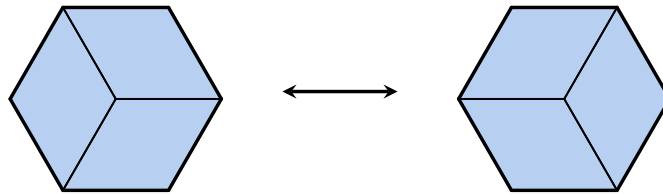


On every board in this packet, each block must cover whole small triangles of the board. Blocks may not overlap or stick out past the thick outline.

Problem 1: Cover the board below with blue rhombuses . Find all the different ways to do it, and draw each way on one of the small boards. Explain how you know you have found them all.

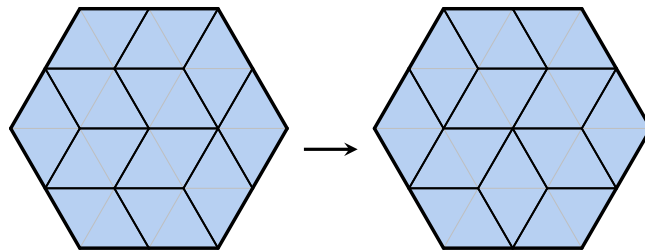
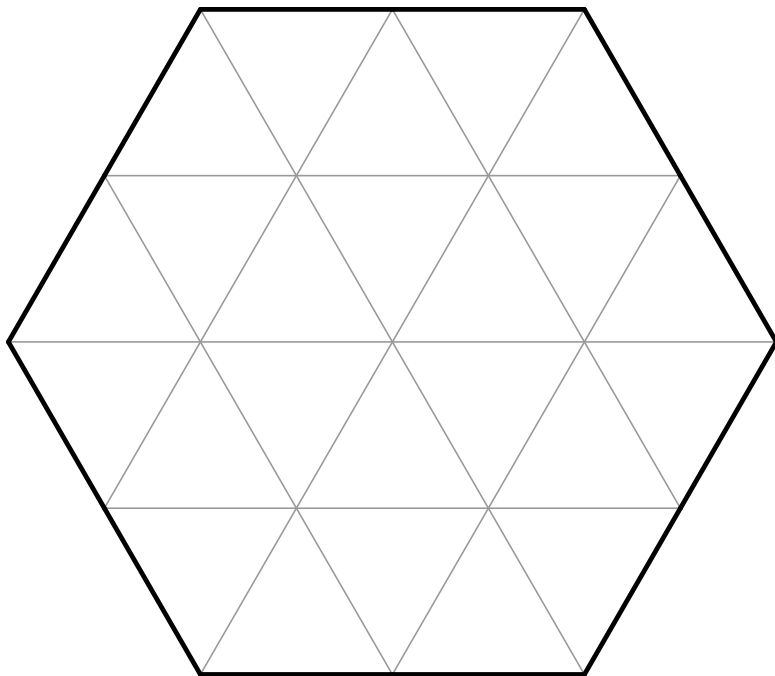


Problem 2: When three blue rhombuses in a covering make a hexagon the size of a yellow block , you can lift them off and put them back the other way, as in the picture. This move is called a flip.



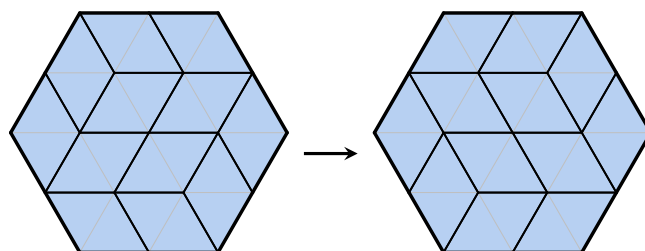
Give each of your coverings in Problem 1 a letter. Below, write the letters and draw a line between two letters whenever one flip changes one covering into the other. Which two coverings need the most flips to get from one to the other?

Problem 3: Put blue rhombuses on the hexagon board to make covering A. Change it into covering B using only flips, with as few flips as you can. Do the same for C and D, and for E and F. Write the fewest number of flips you found in each box.



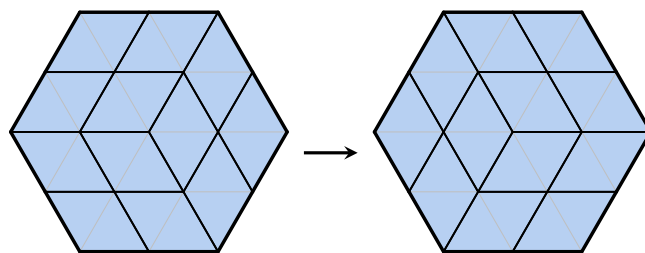
A

B



C

D

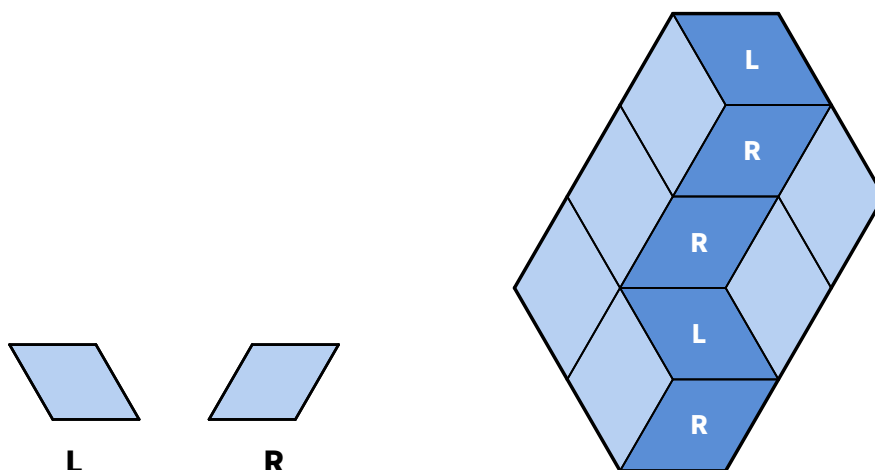


E

F

Problem 4: Start with covering A from Problem 3. Make flips, one at a time, until you are back at the covering you started with. Find a way to do this with 4 flips that never flips the same hexagon twice in a row. Can you do it with 5 flips? With 7 flips?

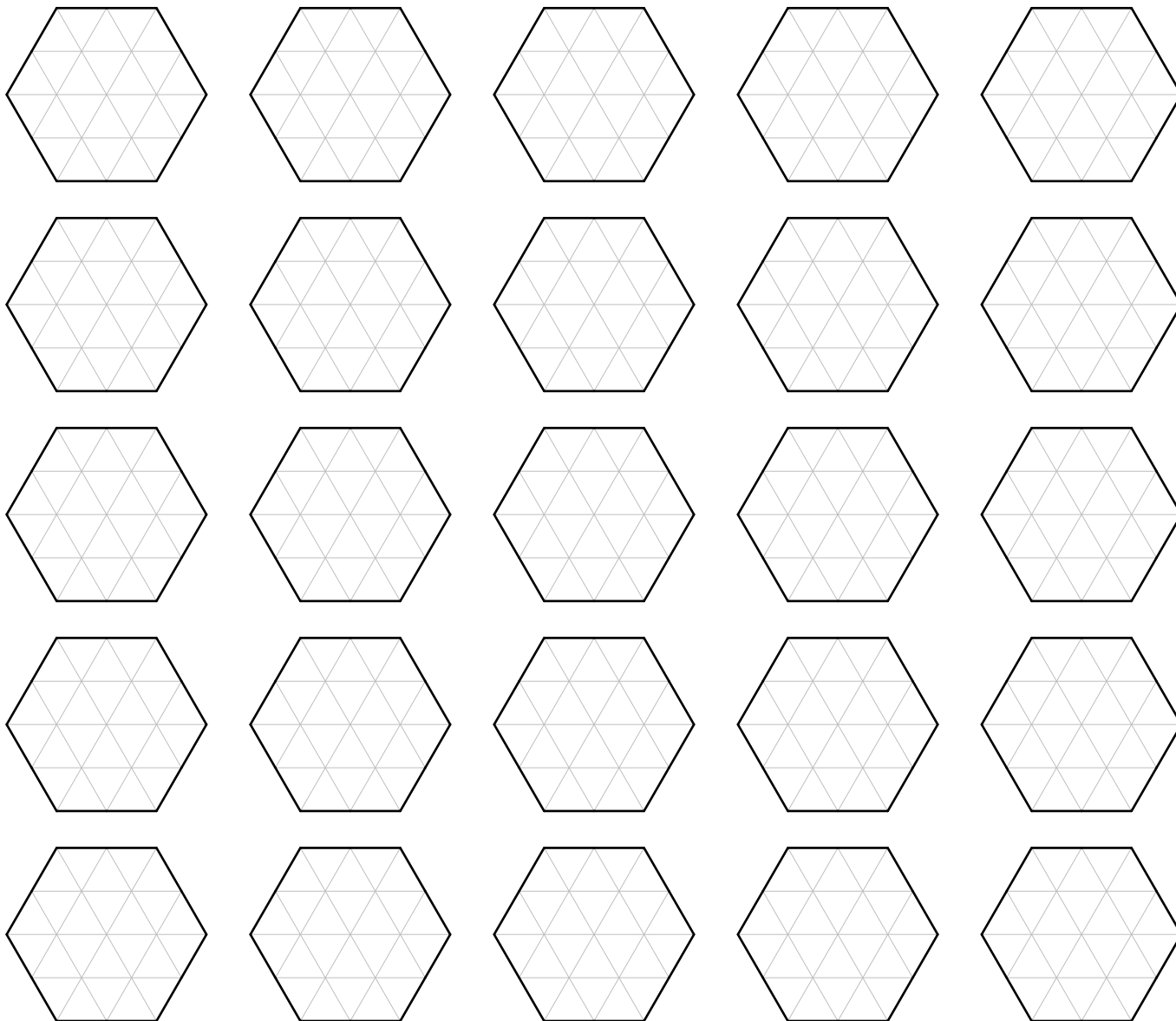
Problem 5: In any covering, pick a short edge on the top side of the board. The rhombus below that edge has a flat bottom edge. Go down to the rhombus below that flat edge, and keep going down this way until you reach the bottom side of the board. Write L for each rhombus you pass through that leans left and R for each one that leans right. The letters, from top to bottom, are the ribbon that starts at that edge. The picture shows a covering of a different board with its ribbon LRRLR.



Write the ribbon of each covering of the board from Problem 1. On the hexagon board each covering has two ribbons, because its top side has two short edges. Find exactly what one flip does to the ribbons.

Problem 6: Explain why, whenever you make flips and end at the covering you started with, you have made an even number of flips. Then explain why nobody can change covering E into covering F in Problem 3 with fewer flips than your answer.

Problem 7: How many different coverings does the hexagon board from Problem 3 have? Draw them on the small hexagons below, and explain how you know you have found them all.



Problem 8: How many different ways can blue rhombuses  cover this board? Explain how you know.

